MILITARY SPECIFICATION SHEET

ELECTRON TUBES, CATHODE RAY

TYPES 5SP1A, 5SP7A, AND 5SP11A[1/]

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Electrostatic deflection and focus with dual gun

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Type phosphor	Ef	Ec1	ed	Eb1	Eb2	Eb3	Rg	Zd	Eb3/Eb2	Alt
Unit:		V	Vdc	v	Vdc	Vdc	Vdc	Meg	Meg	Ratio	ft
Maximum:	All	6.9	0	600	1,100	2,750	8,250	1.5	1.0	3	50,000
Minimum:	All	5.7	-200	---	---	1,000	1,000	---	---	---	---
TEST CONDITIONS:	P1A, P11A	6.3	Adj	---	Focus	1,500	3,000	---	---	---	---
(See note 2)	P7A	6.3	Adj	---	Focus	2,000	4,000	---	---	---	---

GENERAL:

Qualification - Required

[1/] See note 10

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection								
1141	Pressure (implosion)	All		---	---	---	---	---	---
1002	Barometric pressure, reduced	All		---	---	---	---	---	---
5111	Vibration	All		---	---	Width	---	1	mm
1331	Direct-interelectrode capacitance	All	k to all	---	---	Ck	---	5.0	pF
			g1 to all	---	---	Cg1	---	6.6	pF
			D1 to D2	---	---	C1D2	---	2.0	pF
			D3 to D4	---	---	C3D4	---	2.3	pF
			D1 to all except D2	---	---	C	---	2.3	pF
			D2 to all except D1	---	---	C	---	3.0	pF
			D3 to all except D4	---	---	C	---	3.0	pF
			D4 to all except D3	---	---	C	---	2.8	pF
5101	Neck and bulb alinement (electrostatic types)	All		---	---	Dia	---	3.5	Inch
5216	Cathode illumination	All		---	---	---	---	---	---
5246	Focusing voltage (zero-bias)	P1A, P11A		---	---	Eb1	272	521	Vdc
		P7A		---	---	Eb1	363	695	Vdc
5248	Deflection factor	P1A, P11A	1D2; Eb3 = Eb2 = 1,500 Vdc	---	---	DF	49	61	Vdc in.
		P7A	1D2; Eb3 = Eb2 = 2,000 Vdc	---	---	DF	65	81	Vdc in.
5248	Deflection factor	P1A, P11A	3D4; Eb3 = Eb2 = 1,500 Vdc	---	---	DF	43	53	Vdc in.
		P7A	3D4; Eb3 = Eb2 = 2,000 Vdc	---	---	DF	57	71	Vdc in.
5248	Deflection-factor uniformity	All		---	---	---	---	2	%
(C) 1216	Base material insulating quality	All	Zone 5, or higher	---	---	---	---	---	---
	Quality conformance inspection, part 1								
5201	Voltage breakdown	All		---	---	---	---	---	---
5201	Voltage breakdown (electrostatic types)	All		---	---	---	---	---	---
5206	Gas "cross"	P1A	Light = 10 ftL (see note 1)	---	---	---	---	---	---
		P7A	Ib3 = 25 µAdc (see note 1)	---	---	---	---	---	---
		P11A	Light = 4.5 ftL (see notes 1 and 3)	---	---	---	---	---	---
5106	Bulb, screen, and faceplate quality	All		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1 - Continued								
(C) 5223	Modulation	P1A P7A P11A	Light = 10 ftL Ib3 = 25 µAdc Light = 4.5 ftL (see note 3)	--- --- ---	--- --- ---	ΔEc ΔEc ΔEc	--- --- ---	32 42 32	Vdc Vdc Vdc
5221	Light output	P1A P11A	See note 1 See notes 1 and 3	--- ---	--- ---	Light Light	10 4.5	--- ---	ftL ftL
5231	Spot position (electrostatic deflection)	All	See note 5	---	---	---	---	16	mm
5231	Spot displacement (leakage)	All		---	---	Displ	---	10	mm
5241	Grid-cutoff voltage	P1A, P11A P7A		--- ---	--- ---	Ec1 Ec1	-34 -45	-56 -75	Vdc Vdc
5103	Pattern distortion	All	See note 7	---	---	---	---	---	---
5251	Grid No. 1 leakage current	All		---	---	---	---	---	---
5251	Anode No. 2 leakage current	All		---	---	---	---	---	---
	Quality conformance inspection, part 2								
(C) 1101	Secureness of base, cap, or insert	All		---	---	---	---	---	---
1301	Heater current	All		---	---	If	540	660	mA
5201	Electrode current (anode No. 1)	All	Ec1 = 0	---	---	Ib1	-50	10	µAdc
5201	Electrode current (cathode)	P1A P7A P11A	Light = 10 ftL Ib3 = 25 µAdc Light = 4.5 ftL (see note 3)	--- --- ---	--- --- ---	Ik Ik Ik	--- --- ---	1,000 1,000 1,000	µAdc µAdc µAdc
5101	Base alinement (electrostatic types)	All	+4D3, pin No. 4	---	---	---	---	---	---
5101	Side terminal alinement (electrostatic types)	All	-4D3	---	---	---	---	---	---
5101	Angle between traces	All		---	---	---	---	---	---
---	Trace alinement	All	See note 4	---	---	---	---	---	---
5101	Neck and base alinement (electrostatic types)	All		---	---	---	---	---	---
(C) 1111	Base pin solder depth	All		---	---	---	---	---	---
5216	Stray light emission (conventional types)	All	Eb2 = 2,750 Vdc; Eb3 = 8,250 Vdc	---	---	---	---	---	---
5221	Screen	P7A		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 - Continued								
5226	Line width "A" (electrostatic deflection)	P1A	Light = 10 ftL	---	---	Width	---	0.69	mm
		P7A	Ib3 = 25 μAdc	---	---	Width	---	0.69	mm
		P11A	Light = 4.5 ftL (see note 3)	---	---	Width	---	0.69	mm
5226	Line width "B" (electrostatic deflection)	P1A	Light = 10 ftL	---	---	Width	---	0.69	mm
		P7A	Ib3 = 25 μAdc	---	---	Width	---	0.69	mm
		P11A	Light = 4.5 ftL (see note 3)	---	---	Width	---	0.69	mm
5246	Focusing voltage at cutoff	P1A, P11A		---	---	Eb1	345	518	Vdc
		P7A		---	---	Eb1	460	690	Vdc
5248	Deflection factor	P1A, P11A	1D2	---	---	DF	62	76	Vdc/in.
		P7A	1D2	---	---	DF	83	101	Vdc/in.
5248	Deflection factor	P1A, P11A	3D4	---	---	DF	53	65	Vdc/in.
		P7A	3D4	---	---	DF	70	86	Vdc/in.
---	Deflection factor ratio	All	See note 6	---	---	---	---	---	---
---	Interaction factor	All	See note 8	---	---	---	---	---	---
5251	Heater-cathode leakage current	All		---	---	---	---	---	---
1105	Permanence of marking	All			---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	All	Group C; Ib3 = 15 μAdc; Eb2 = 2,750 Vdc; Eb3 = 8,250 Vdc	---	---	t	500	---	hrs
---	Life-test end points:	P1A	Light = 7.5 ftL	---	---	---	---	---	---
		P11A	Light = 3.5 ftL (see note 3)	---	---	---	---	---	---
		P7A	Ib3 = 20 μAdc	---	---	---	---	---	---
	Modulation	P1A, P11A		---	---	ΔEc1	---	32	Vdc
	Modulation	P7A		---	---	ΔEc1	---	42	Vdc
	Line width "A"	All		---	---	Width	---	0.69	mm
	Line width "B"	All		---	---	Width	---	0.69	mm

NOTES:

1. This test to be performed at the conclusion of the holding period.
2. All electrical tests are to be made on each gun separately.
3. As measured with a 2 inch x 2 inch (50.8 x 50.8 mm) raster using type 3 photronic cell without eye correction, calibrated in footcandles of illumination from a light source having a color temperature of 2,700°K.

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NOTES: - Continued

4. Corresponding traces of each gun should be within 1 degree of each other.

5. Under stable operating conditions (method 5231), the position of either spot will not shift with changes in intensity by more than 0.025 inch (0.64 mm).

6. Ratio of deflection factors of corresponding sets of deflection plates (larger deflection factor divided by lower deflection factor) shall not exceed 1.15.

7. With a raster pattern the size of which is adjusted so that the widest points of the pattern just touch the sides of a square, 3.075 inches (78.11 mm) on a side, no point on these pattern sides will lie within an inscribed square, 2.925 inches (74.30 mm) on a side.

8. The deflection of one beam when balanced dc voltages are applied to the deflection electrodes of the other beam will not be greater than the indicated value.

9. Maximum voltages applied to base, deflection plates and A2 only.

10. Tube type 5SP2A has been deleted from this specification sheet.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11
DSA - ES

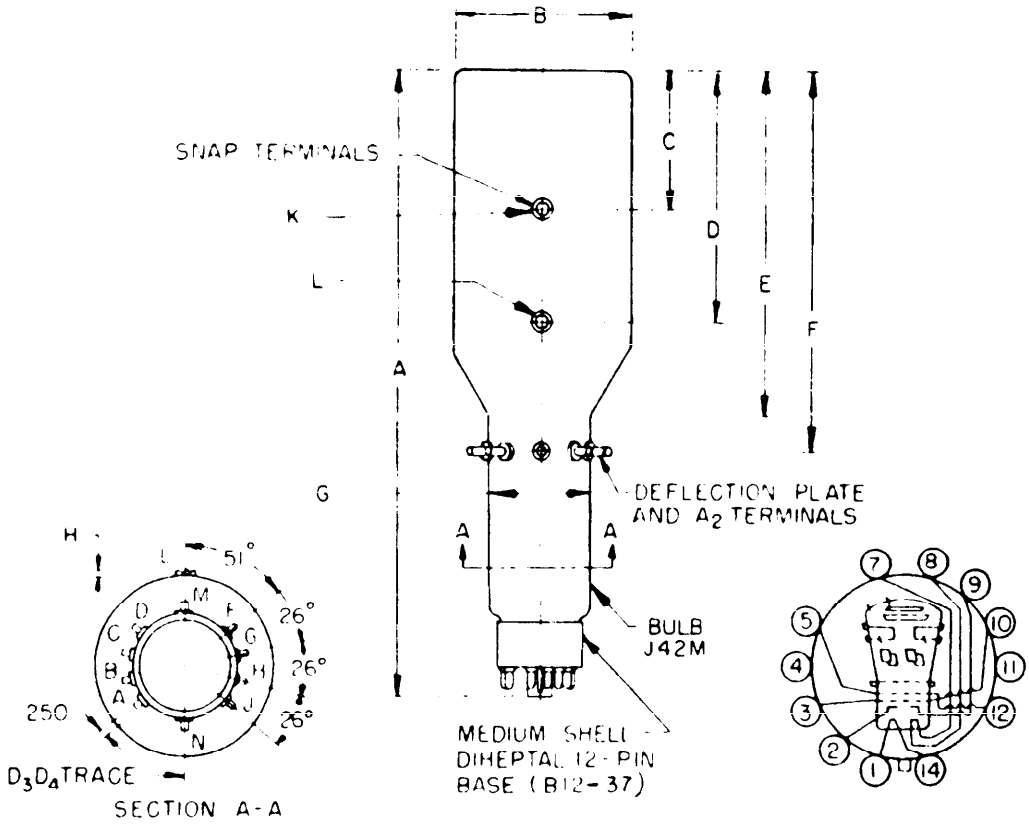
User activities:
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Navy - AS, OS, MC, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2554)

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Connections

Terminal K Intensifier Electrode
Terminal L Accelerating Electrode Coating

Unit A		Unit B	
Pin No. 1.	Heater	Pin No. 7.	Heater
2.	Cathode	8.	Heater
3.	Control Electrode	9.	Cathode
4.	Internal connection	10.	Control Electrode
5.	Focusing Electrode	11.	No connection
14.	Heater	12.	Focusing Electrode
Terminal A	Deflection Plate D2	Terminal F	Deflection Plate D1
B	Deflection Plate D1	G	Deflection Plate D2
C	Deflection Plate D3	H	Deflection Plate D4
D	Deflection Plate D4	J	Deflection Plate D3
M	Accelerating Electrode	N	Accelerating Electrode

FIGURE 1. Outline drawing of electron tube types 5SP1A, 5SP7A, and 5SP11A.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	18.000(457.20)	18.500(469.90)
B	5.156(130.96)	5.344(135.74)
C	3.750 (95.25)	4.250(107.95)
D	7.000(177.80)	7.500(190.50)
F	10.750(273.05)	11.250(285.75)
G	2.937 (74.60)	3.062 (77.77)
Quality conformance inspection, part 3 (periodic check)		
BULB: J42M		
BASE: B12-37		
Reference dimensions (see note c)		
E	10.000(254.00)	
H	.125 (3.18) MAX	

NOTES:
a. M-N connected together internally.
b. L and M-N shall be connected together to a common ground.
c. Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube types 5SP1A, 5SP7A, and 5SP11A - Continued.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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